IMPORTANCE MASKING & MULTIOBJECTIVE MLM FOR IDR PREDICTION

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ABSTRACT

We present our work on predicting intrinsically disordered regions (IDRs) using deep learning models that leverage both fine-tuned and non-fine-tuned embeddings, as well as aggregated versus non-aggregated sequence representations. First, we split and curated our training data by clustering protein sequences with mmseqs at 30% sequence identity and then refining the splits to improve divergence metrics. We hypothesized that human proteins with binding activities would serve as an effective source of fine-tuning data, so we obtained 13,000 sequences from UniProt and integrated mutation scores, predicted binding probabilities, and disorder confidences into custom per-residue vectors. In addition, we introduced a difficulty-weighted masking approach during fine-tuning, where masked language modeling (MLM) targeted residues with high IDR ambiguity, elevated binding complexity, or critical mutation severity rather than relying on random masking, thereby sharpening the model's attention to the hardest residues. To evaluate our models, we used four primary metrics—precision, recall, F1-score, and AUROC—placing special emphasis on reducing false negatives, which are costly in disorder prediction. Our experiments revealed that a non-aggregated approach with fine-tuned embeddings achieved the highest performance, indicating that preserving residue-level information and tailoring embeddings to IDRs offers tangible benefits despite increased computational demands. We also tested aggregated embeddings, which were more efficient to train but lost some local details essential for pinpointing disordered residues. In conclusion, we show that specialized data, particularly human proteins with experimentally verified binding sites, can meaningfully enhance disorder prediction. Although computational limita-

tions and data availability remain our primary constraints, our findings underscore the value of tailoring language model embeddings toward sequence-specific tasks and highlight pathways for improving structure prediction in biologically complex proteins.

1 INTRODUCTION

Proteins play a crucial role in maintaining life, being responsible for a wide variety of biological processes ranging from gene expression to energy production and muscle contraction [1]. With proteins being so important and spanning a vast range of applications, the structure of proteins are pertinent in defining their biological function and interactions with other systems. Their function is particularly impacted by the three-dimensional structure of the protein which is determined by the amino acid sequence, and slight alterations in structure can lead to disorder and mutations leading to diseases [2]. As such, predicting protein structure has many applications in drug discovery, examining genetic disorders, vaccine development, and general protein design. Deep learning has enabled the evolution of structure prediction by integrating computational approaches with bio informatics. This approach is driven by the ability to quickly learn patterns from large datasets of known protein structures, facilitating highly accurate predictions of how sequences fold into 3D structures. Models like AlphaFold implemented a transformer based architecture allowing the spatial and evolutionary relationships between amino acids to be stored, leading to the development of the AlphaFold protein structure database which provided predictions for most of the human proteome [3]. Other models like ESMfold implemented ESM-2s large scale protein language model which facilitated the prediction of structure without needing multiple sequence alignment [4].

Intrinsically disordered proteins (IDPs), and intrinsically disordered regions(IDRs) cause a lack of both secondary and tertiary structure leading to their involvement in many diseases. Although they play a crucial role in various cellular processes like cell signaling and regulation, being very important in biology, little is known about their precise function mainly because their lack of structure makes them difficult to measure [5]. Given the importance of understanding disorders in protein analysis, predicting disordered regions is a paramount goal of deep learning models. Despite improvements in sequence prediction with the integration of deep learning, there are still several obstacles in predicting disordered regions. One of the main challenges is the lack of experimental data for disabled sequences. Many models, like AlphaFold, need experimental data to make a quality training set and compare the results of the model to, so lacking experimental data makes hinders the application of deep learning in disordered predictions. Another issue is the lack of structure in disordered regions, which is especially troubling to models that use information about protein structure in their embeddings and ultimately sequence prediction.

The two primary characteristics changed between our prediction models were fine-tuned vs non-fine tuned embeddings, along with aggregated vs non aggregated sequence embeddings. Fine-tuning was a very powerful tool throughout our project since it enabled task-specific optimization of our embeddings, which allowed for better performance in predicting disorder than the standard pre-trained non-finetuned esm embeddings that weren't specialized for disorder prediction. One downside in finetuning was the computational resources it took, and the amount of data we needed to procure in an attempt to counter overfitting. Aggregating the embeddings allowed for fixed input layers reducing the complexity of the model, and was also good at capturing global sequence features. However, aggregation also resulted in a loss of some residue level information that did not happen in non aggregated embedding systems. Non aggregated sequences, while preserving more information, led to more complex model designs and longer training times. During part D, we found non-aggregated sequences that were embedded using a fine-tuned model to be the most successful, because we were able to preserve residue level while employing a specialized embedding model.

2 METHODS

2.1 CURATING PROVIDED DATA

We were provided labeled training and testing sets containing sequences of length L, along with corresponding disorder vectors of length L. The training set included 4,751 sequences, while the

test set included 573 sequences. The initial portion of data curation was centered around the separation of the given training set into a training and validation set, which was done by implementing clustering and KL divergence testing for disorder sequence similarity. The first step was clustering the entire dataset, which was done by applying mmseqs classic clustering with a minimum sequence identity of 30%, minimum alignment coverage of 50%, default coverage mode, and a cluster mode of 3 which uses single-linkage clustering applying a transitive property between sequences to make larger clusters. Following clustering, we applied an 80-20 training validation split to the clusters to ultimately end with a training set of 3,815 sequences and a validation set of 936 sequences. After clustering, we observed a unique issue in the data split due to a similarity in the disorder vectors between sets. This was because many of the sequences experienced disordered groups in the beginning and end of the vectors with non disordered stretches in the middle, which would influence the model to memorize this pattern instead of learning as intended. We calculated the KL divergence between training/test and validation/test sets which scored .0045 and .0049 respectively meaning that on average the training disorder was extremely similar to our test set. After sorting each sequence by KL divergence we restructured the splits such that the divergence between validation and test set rose to .177, while the divergence between training and validation set rose to .315, Ultimately allowing the sets to be separated by a combination of unique clusters and disorder vectors so that the model was not overfit.

2.2 Human Binding Data for Fine-tuning

2.2.1 Importance Masking

As IDRs are extensively involved in protein binding activities to different substrates [citation], we hypothesized that fine-tuning based on human proteins with binding activities will provide a better representation of IDR-related properties for our model. We downloaded all manually reviewed human protein sequences associated with the umbrella Gene Ontology Molecular Functions (GO:MF) term, GO:0005488 binding, from the UniProt database [6]. After acquiring the initial dataset we sought to make vector representations of each sequence that we could specialize our encoder with through fine-tuning. The vector representations were generated by first searching for variation data corresponding to each sequence in our dataset. A vector of 0s of the same length of each sequence was created, and then whenever there was a variation, a scaled value was placed in the vector at the corresponding position of the variation based on whether it caused deleterious or benign mutations.

To implement a importance-based masking strategy, we leveraged SaLT&PepPr and AIUPred to evaluate the likelihood of each residue participating in protein-protein interaction and IDR formation [7]. The confidence of binding activity and IDR prediction per residue was calculated using equation (1), with a prediction of 0.5 being the least confident and a prediction of 0 or 1 being the most confident. The per-residue prediction was also aggregated into an uncertainty score for the whole sequence, indicating the overall difficulty through equation (2), which gives sequences with more low-confidence residues higher uncertainty scores.

$$\text{Confidence} = 2 \times |0.5 - \text{Prediction}| \tag{1}$$

$$\text{Uncertainty} = \frac{\sum(2 \times (0.5 - |0.5 - \text{Prediction}_n|))^2}{n} \tag{2}$$

Instead of random masking, we implemented a difficulty-based masking strategy for the fine-tuning of the ESM2 model for the task of IDR prediction. We prioritized the masking of residues that were estimated to be more challenging for the model to predict, or relevant to the task. The difficulty of individual residues was calculated based on the aforementioned IDR model and binding prediction confidence generated with SaLT&PepPr and AIUPred. Previous studies have reported that missense mutations from IDRs are usually benign except for unique cases where the IDRs are involved in phase separation. Others have found protein-expansion is primarily driven by indels in IDR regions [8]. Thus, we have additionally incorporated the mutation data from UniProt for prioritizing the masking of residues with benign mutations under the assumption that residues with more benign mutations are more likely to come from IDRs. With this masking strategy, we ensured the fine-tuned model was specifically trained to predict residues that are of particular importance to IDRs.

We also consider curriculum masking, varying masking percentage via a cosine scheduler from a lower bound to an upper bound over the course of each epoch. We used 15% to 45%, following

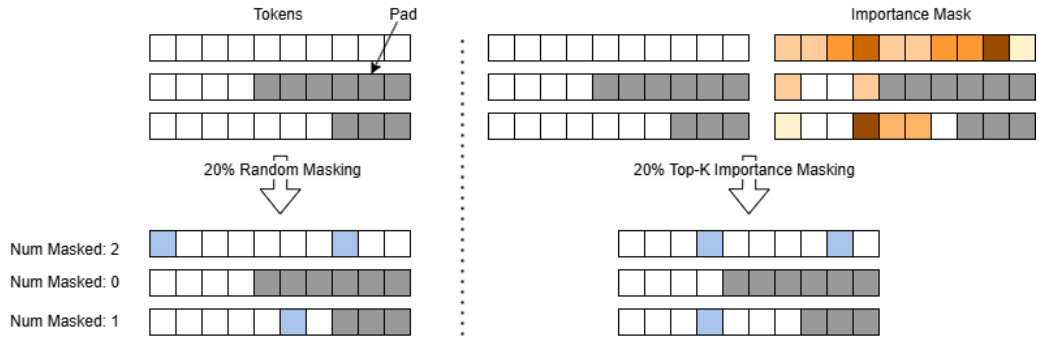


Figure 1: Importance masking example.

intuitions from ESM-3 and other similar papers. We considered this in conjunction with importance masking, as can be seen in Table 1 and 2. For simplicity, we call this 'curricula MLM'.

2.3 Additional Training Objectives

Prior work has found IDRs to be highly contextual, displaying several states conditional on the current cellular environment - as opposed to being static ensembles linked by disordered regions [9]. Thus we reasoned that incorporating more information about the context the protein is typically found and operates in would improve representations with respect to disorder. To do so, we extracted the Gene Ontology IDs associated with each protein from UniProt, each encoding fine-grained information about the protein (ex: 'heme binding'). We then applied the Map2Slim tool from OWLtools to map this information onto slimmed subsets, then converting these subsets into three standard boolean vectors - encoding molecular function, biological process, and subcellular localization respectively [10]. This output largely matches the UniProt implementation of GO annotation vectors, as seen in Fig. 2, though we note this information is not directly present in the database. These incorporated into the model as an additional training objective during MLM fine-tuning, via an element-wise binary cross entropy loss (not a multiclass, as multiple classes are often true simultaneously).

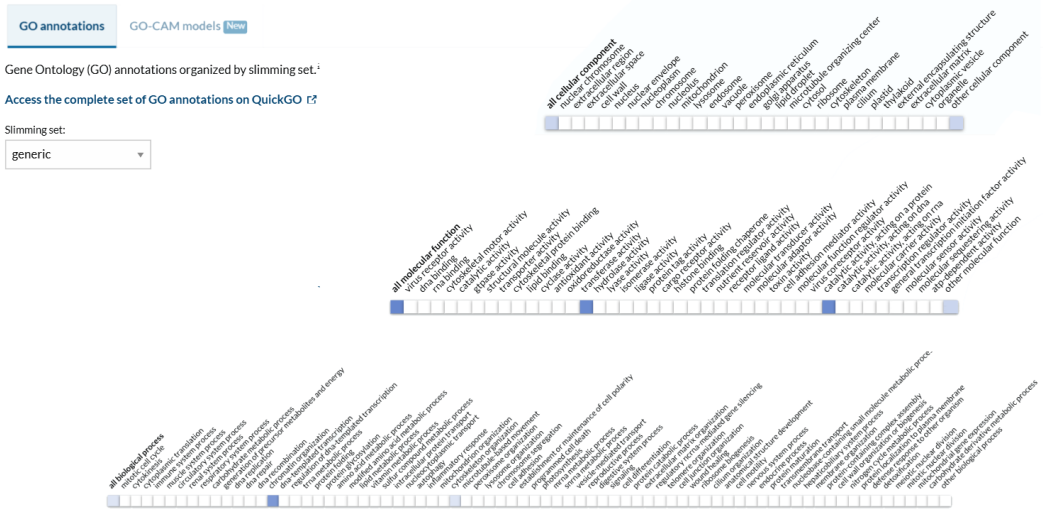


Figure 2: UniProt GO annotation example, for protein with accession A0A0A0MRJ0.

2.4 EMBEDDINGS

We started with pre-trained ESM2 embeddings (pretrained.esm2-t6-8M-UR50D) for the model, which had an embedding length of 320 per residue. For the all the models we initially created matrices for each sequence that had L(length of sequence) rows and 320 columns. For the aggregated models we took the mean across the matrix such that we flattened it into a 1 dimensional vector representation; for the non aggregated models we kept the matrices as is such that we could keep all residue information when producing a prediction for disorder at each position. We went with a Figure 3: Classification head architecture diagram. Energy module is an unpacked version of IUPred weightings, a classical energy based method for predicting energy, which can be used to predict disorder. Model B and D have the same diagram besides the fine tuning block being toggled embedding approach for our fine-tuned data since simply creating more esm embeddings wouldn't help specialize our model for the given task. As discussed in section 2.2, we used human proteins that had active binding sites for our fine-tuning set. Vectors were generated of equal length to the sequence, and at any position that a mutation occurred a scaled combination of the polyphen and sift score was added which represented how damaging or benign the mutation was.

2.5 ARCHITECTURES

Our model was designed with ablation in mind, such that fine-tuning and aggregation could be toggled in order to easily determine their impacts on performance. There were 2 primary architectures defining the model, with the first being related to fine-tuning, and the second being the implementation of the classification head.

2.5.1 FINETUNING MODEL

We used ESM-2 8M for our fine-tuned model for simplicity (ex: easy access to attention intermediates) and speed of debugging. We unfroze the last 2 of 8 attention layers and language head of the model during fine-tuning, and pruned sequences to 2400 to avoid quadratic memory costs from attention (only around 3% of 13k dataset was above these lengths). We selected small batch sizes for similar reasons. We trained for 10 epochs with early stopping, which was common near the end of training.

Masking was done on per protein basis rather than over the batch for more stable training dynamics (see Fig. 1 for example).

GO annotations were predicted via a classification head on the CLS token and a lightweight few-head attention module over the residues - with the idea behind the latter being to promote residue-specific contextual information, such as a motif that indicates cellular location.

2.5.2 CLASSIFICATION HEAD

ESM-2 describes a method for unsupervised contact prediction involving a simple logistic regression layer over a symmetricized and mean-centered version of the final model attention map [4]. Our classification head used an attention layer on top of the hidden state output of the ESM-2 model. Additionally we applied gated embedding masking, which was inspired by the PTM-Mamba model, which combined the embeddings from ESM-2 with the output of the classifier dynamically by learning a new set of weights for this function. This was very useful since the original ESM-2 embeddings contained structural features that were preserved through the classification head. Overall, we applied a multiple-faceted feature extraction strategy to the protein sequence. We combined the energy embedding generated from an unpacked version of IUPred weightings, a learned merged embedding generated from SaLT&PepPr and ESM-2, and the "nearby attention" vector generated from the original ESM-2 embedding for the multi-head self-attention (MHSA) layer, enabling the model to learn the complex relationships between these features for the prediction of IDRs. A sigmoid activation function was applied after the MHSA layer to generate the binary classification.

$$\text{APC}(A)_{ijk} = A_{ijk} - \frac{\text{mean}(A_{ij}) \cdot \text{mean}(A_{kj})}{\text{mean}(A)}, \text{Symm}(A) = \frac{A + A^T}{2}$$

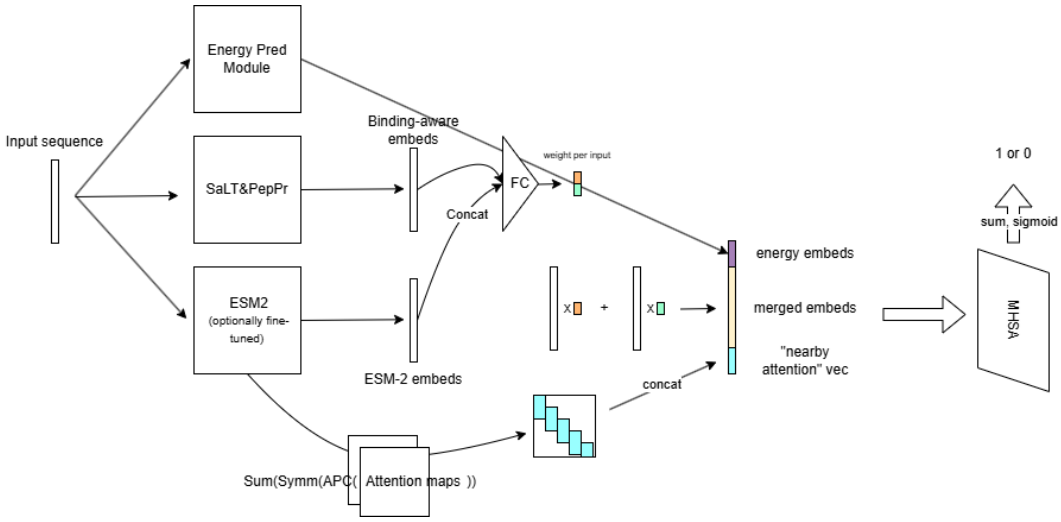


Figure 3: Classification head architecture diagram. Energy module is an unpacked version of IUPred weightings, a classical energy based method for predicting energy, which can be used to predict disorder. **Model B and D have the same diagram besides the fine tuning block being toggled**

Model A was the simplest one and functioned by turning aggregation off, and using a non fine-tuned model just feeding embeddings directly from the pre-trained ESM-2. Model B was also non fine-tuned and was ran by turning aggregation on, while keeping fine-tuning off. Model C was aggregated but implemented a fine tuned model, and model D combined the best fine-tuned model we ran with the best classification head to end with a non-aggregated fine-tuned model.

Energy module is an unpacked version of IUPred weightings, a classical energy based method for predicting energy, which can be used to predict disorder. Model B and D have the same diagram besides the fine tuning block being toggled (see Fig. 3).

2.6 Evaluation Metrics

To evaluate the effectiveness of the model we compared our predicted percentages with the given labels by examining 4 criterion, F1-score, precision, recall, and AUROC. All of these metrics were based on the calculations of true and false positives. Precision measures the amount of predicted positive cases that were actually positive which quantifies the false positive rate of our model. Recall measures how many positive cases were accurately identified. F1-score is the harmonic mean of precision and recall which provides a balanced metric for when both false positive and negatives are important. Finally area under the receiver operating characteristic curve (AUROC) measures the ability of the model to distinguish between positive and negative cases by integrating under the product of the true and false positive rates. While all metrics provide key insight to the performance of the model AUROC and recall are especially important in this context; maximizing recall means we are reducing false negatives and predicting false negatives in the context of disordered regions is disastrous.

3 RESULTS

3.1 FINE-TUNE RESULTS

Model	Perplexity
ESM-2 baseline	2.468
Finetuned baseline	2.915
FT + GO annotations	2.872
FT + Curricula GO	
FT + Curricula MLM	2.136
Importance Masking (I): IDR only	2.994
I + Bind only	2.942
I + Mutability only	2.959
I + Merged IDR+Bind+Mutability	2.938
I + Curricula MLM & Merged	2.139

Table 1: Pseudo-perplexity scores of different models on fine-tune task. Calculated as standard, via masking iteratively over samples in test set. Lower values indicate better performance.

We covered a variety of different fine-tuning techniques, broadly sorted into importance masking and curricula MLM (increasing the masking rate over the course of each epoch). We found that generally perplexity was significantly impacted mostly by curricula MLM, with importance masking either worsening or not improving results. This is expected, as perplexity is calculated in a IDR-independent fashion and we would assume the purpose of importance masking is to tailor the results to IDRs. Interestingly, addition of GO annotations as a task slightly improved performance, despite competing for model parameters with the primary MLM loss function, indicating it improves model representations.

3.2 CLASSIFICATION HEAD RESULTS

The embedding techniques started off very similarly in all models as we created a matrix of ESM-2 embeddings with each row being one amino acid vector with a length of 320. In models A and C we aggregated the embeddings such that they were represented in a single vector instead of matrix, then ran them through the model; Models B and D were non aggregated to preserve residue information.

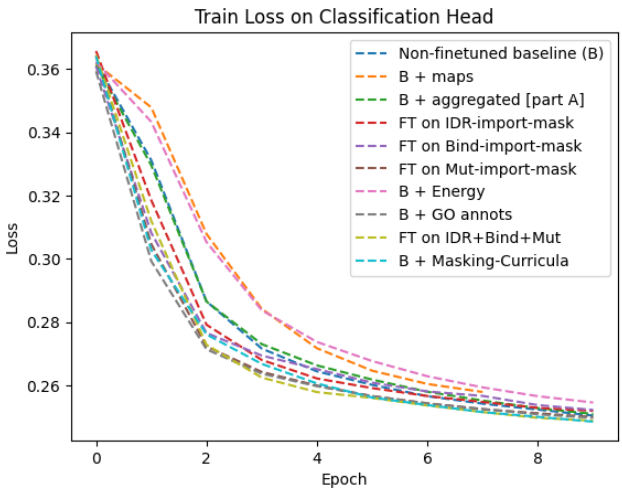


Figure 4: Classification head ablations performance on validation.

We used a variety of classification head approaches, sorted into baseline (non-finetuned) to fine-tuned. All non-Part A approaches created an embedding per residue and passed it through MHSA,

as per Fig 3, sum-aggregating and applying sigmoid to get a score. Overall, most methods did not significantly improve results, with the standout being curricula MLM, where masking rate is increasing (from 15% to 45%) over the course of the batch. Interestingly, this performance boost is retained even in the fine-tuned plus importance masking classification head (last result in Table 2). Otherwise, GO annotations seem slightly positive. Aggregation over residues (Part A result) had predictably poor performance, as the model is forced to predict the same verdict (disordered / not disordered) for every residue.

Model	Precision	Recall	F1-score	AUROC
Non-finetuned baseline [part B]	0.6192	0.3034	0.3564	0.8529
Non-FT + maps [part B]	0.6119	0.2601	0.3171	0.8470
Non-FT + aggregated [part A]	0.3412	0.2128	0.2634	0.6517
Non-FT + Energy [part B]	0.6117	0.2932	0.3456	0.8448
Non-FT + GO annotations [part B]	0.6252	0.3083	0.3632	0.8518
FT + GO-Curricula [part B/D]	0.6172	0.3147	0.3676	0.8561
FT + Masking-Curricula [part B/D]	0.6172	0.3147	0.3676	0.8561
FT on IDR Uncertainty [part C]	0.6138	0.2992	0.3567	0.8468
FT on Bind Uncertainty [part C]	0.6125	0.3103	0.3649	0.8469
FT on Mutability [part C]	0.6112	0.3532	0.3996	0.8553
FT on IDR+Bind+Mutability [part C]	0.6149	0.3233	0.3771	0.8526
FT + IDR+Bind+Mutability + Masking-Curricula [part D]	0.6122	0.3281	0.3767	0.8526

Table 2: Performance metrics of different disorder classification models. Arguably most are Part D, but for simplicity they're mapped to the task where they could be performed first.

4 CONCLUSION

Throughout the process of creating these models, we found that the fine-tuned non-aggregated model provided the best performance. While this model provided the best results, it was still limited by computational power and data collection. Unfreezing a majority of the layers in our fine tuned model demanded a lot of computational power, making fine tuning very time consuming utilizing the GPUs in Google Colab. Even when we increased the GPU runtime allocation, the bottleneck shifted to data I/O overhead, underscoring the need for more advanced hardware solutions. This strain was compounded by not aggregating the sequences since the length of each embedding sequence was now $L \times$ *embedding_dim* rather than just *embedding_dim*. The other limitation was procuring data that could be used for fine-tuning. We implemented a set of 13,000 human proteins with binding activities, which is quite minuscule in comparison to the dataset that the pretrained model was developed on. While we attempted to obtain additional sequences from public repositories, many lacked high-quality annotations, thus limiting their utility for robust fine-tuning. Procuring more data for finetuning would further improve the models performance and reduce overfitting. However increasing finetuning data would also further compound the overall computational load on our model. Looking forward, adopting optimized training loops and exploring advanced fine-tuning techniques—like LoRA or adapter modules—may provide a more resource-conscious path to higher performance.

Overall we did not find that much of a performance benefit from the importance masking, even IDR-guided importance masking, which is surprising. One would expect that a larger focus on the most-difficult-to-predict-as-IDR residues would result in a better understanding of IDR properties. This result potentially suggests a future ablation of directly guiding masking via IDR scores, to determine whether or not the error lies within attempting to guide masking or using model uncertainty as a method to do so. Other explorations might include directly predicting the "benign or pathogenic" mutation vector during MLM as an auxiliary loss ala GO annotations, as those were relatively more successful. Overall, our results suggest a divergence between the underlying intuitions behind MLM and standard biological models of IDR functions, providing an interesting result potentially for future research to expound on.

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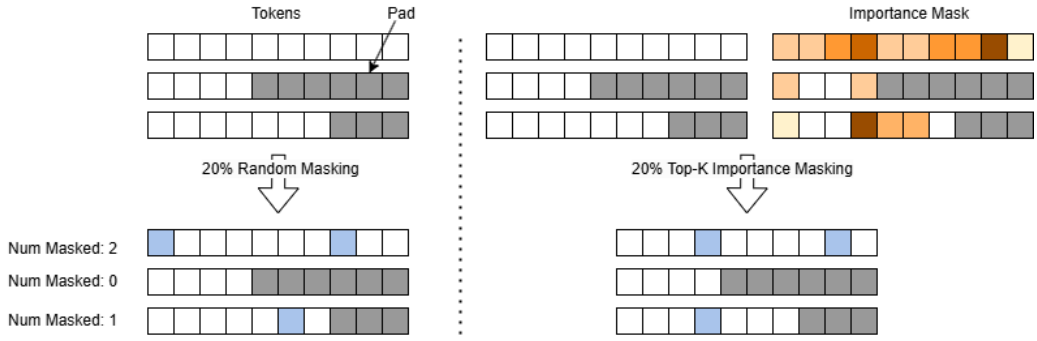


Figure 5: Importance masking example.

Names	[Finetuning] Pseudoperplex	[Classification Head] AUROC (Epoch 1 $\rightarrow$ 10)
Non-FT Baseline (raw ESM2)	2.47	0.659 $\rightarrow$ 0.853
FT Baseline	**2.08**	0.632 $\rightarrow$ 0.851
FT BS+Pred GO:	2.87	0.410 $\rightarrow$ 0.852
Importance Masking on uncert IDR	2.99	0.523 $\rightarrow$ 0.847
IM on uncert S&P	2.94	0.567 $\rightarrow$ 0.847
IM on direct IDR	2.12	0.629 $\rightarrow$ 0.851
IM on direct S&P	2.12	0.660 $\rightarrow$ 0.851
IM on Mutat	2.10	0.632 $\rightarrow$ 0.856
IM on Merged (I+SP+M)	2.14	0.660 $\rightarrow$ 0.841
Curricula on Std Mask	2.14	0.603 $\rightarrow$ 0.851
Curricula on GO	**2.06**	0.662 $\rightarrow$ **0.856**
Curricula on IM Mutat	2.17	0.714 $\rightarrow$ 0.848
Curricula on IM Mutat+GO*	2.11	0.733 $\rightarrow$ 0.845
Non-FT BS+Bind_Embeds (gated)	n/a	0.754 $\rightarrow$ 0.853

Table 3: Comparison of different finetunings of ESM-2 8M on 13k human binding dataset. Epoch 1 = after epoch 1 on training. Epoch 10 = final training perf. on test. Early stopping. *Interleaved